

UNIT 6

MINERAL RESOURCES

A mineral is defined as being a naturally occurring element or compound that is formed by inorganic processes and contains a crystalline structure.

Geographers are primarily concerned with minerals in soil because minerals form the basic framework of soil.

Minerals originally form when **once-heated Earth material magma** (molten rock) cools and forms solid igneous rock. The Earth's crust formed and continues to form in this manner.

Earth's crust contains a combination of naturally occurring elements, of which following elements are predominant: silicon, aluminum, iron, calcium, Sodium, potassium, and magnesium. As you can imagine, combinations of these elements along with the other naturally occurring elements that form Earth's crust produce a wide variety of minerals.

Igneous rocks contain original minerals that form as magma cools but **sedimentary rocks** are formed by secondary minerals that grow and join sediment particles together and become cemented. **Metamorphic rocks** were once igneous rocks and sedimentary rocks that become chemically altered to form different minerals.

Key Points:

Formation of minerals:

- Over 3000 minerals are currently known and about 50 new discovered each year.
- Some minerals are originally formed from hot **magma**, which contains the minerals. When the magma cools crystals of minerals appear.
- Most of the minerals are formed underground when **heat and pressure** transform one form of rock into another.
- **Decomposition** of leaves, plants and bones, flesh ultimately transform into minerals but it takes million of years.
- Mostly this process of decomposition happens in oceans where thousands of species dies every day.

Mining processes:

- Mining is a process of digging rocks and minerals from the earth.
- Minerals are found at different depths.
- There are three main methods of mining.

(a) Open cast mining:

- Some minerals like coal and iron often lie near the surface.
- Open cast mining scoops up these minerals from near the surface with the help of giant excavators and power shovels.
- Which then load the material into Lorries or railway wagons to be carried away.

(b) Under ground Mining:

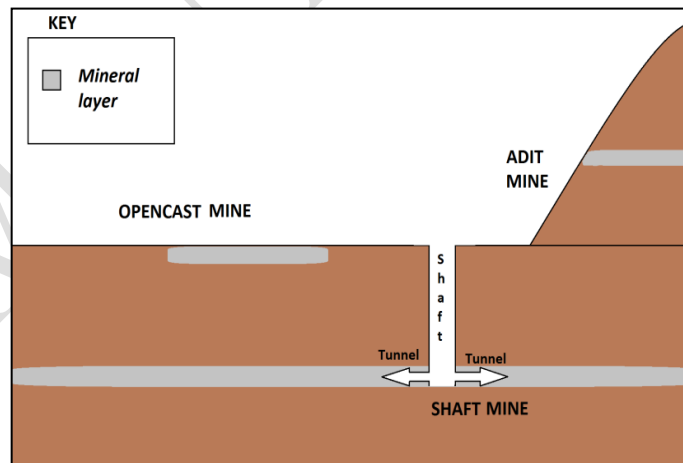
- There are two methods by which minerals are accessed underground.

(i) Adit mining:

- An adit is an opening or passage.
- Adit mining is done in hilly districts where a mineral seam is exposed on a hill side.
- Horizontal tunnels are dug into the side of a valley or hill to reach the mineral deposit.

(ii) Shaft mining:

- Vertical shafts are dug down to the minerals, especially for coal.
- This method is expensive and can be dangerous.



Metallic and Non metallic minerals:

- Metallic and non metallic are one of the mankind's most highly prized possessions.
- Many of our articles are made from metallic minerals.
- Few elements such as gold and copper occur in pure form as "metallic minerals", but most are found as "ores".
- **Ores** are compounds containing a high proportion of the metal.

- These metallic ores are cut or blasted from surrounding rock. The ore is crushed and the worthless rock removed.
- Mineral resources are non renewable although many can be recycled and used again.

Metallic minerals are:

- Iron ore
- Copper antimony
- Chromites
- Celestite
- Manganese
- Gold
- Silver
- Tin
- Bauxite

Non Metallic minerals are:

- Coal
- Sulphur
- Rock salt
- Gypsum
- Soapstone
- Limestone
- Marble
- Clays

Organizations for mining in Pakistan:

- ***Geological Survey of Pakistan***, started working in 1947.its main function was to investigate the minerals deposits.
- ***Pakistan Mineral Corporation*** started working in 1974.its main function was to explore and market all the minerals. The **Pakistan Mineral Development Corporation** (PMDC) is an autonomous corporation attached to the Ministry of Petroleum and Natural Resources, of the Government of Pakistan.
- ***Resource Development Corporation*** started working in 1974.it used to investigate and develop copper mines at Saindak, Balochistan.
- ***Gemstone Corporation of Pakistan*** established in 1979.its main aim was to develop gemstone resources.

Description and uses of metallic and non metallic minerals:

(i) Rock salt:

- Seams of rock salt vary in thickness from between 20 to 100 meters.
- The rocks are whit and pink in color.
- The salt is overlain by gypsum and clay.

- Rock salt is used for cooking and preservation purposes and for the manufacture of soda ash, caustic soda and other sodas for laundries, textiles and tanning.

(ii) Brine:

- Used in chemical and fertilizer industry.

(iii) Limestone

- Limestone is a major sedimentary deposit and is widespread in Pakistan.
- It is the main raw material for cement.
- It is also used in the manufacture of bleaching powder, glass, soap, paper, paints and lime.
- It is used to treat sugarcane waste to produce alcohol fuel.
- It is painted on barks of trees to counter pests and termite attacks.

(iv) Coal

- Pakistan has low quality coal is mainly used in brick kilns.
- Some is use to make coal.
- A small percentage is used for power generation.
- Thermal power stations are also being run by coal.

(v) Natural gas;

- Gas is being used in Domestic and industrial uses.

(vi) Gypsum:

- Found is grey, white and pink color.
- It is used in the manufacture of paints, fertilizers and pre – fabricated construction boards.
- White gypsum is used for making cement and plaster of Paris.
- Spread on saline soil to help land reclamation for farming.

(vii) Marble:

- Found in bands of white, grey, yellow and brown.
- It is used in buildings and for making chips for flooring and decorative pieces.

(viii) Clays

- China clay is used in the ceramic industry, for a special type of cement and has other industrial uses.
- Fire clays fine clay capable of enduring high temperature to make fire bricks; it is also used to make pottery and chemicals.

(ix) Magnetite

- It is used in the manufacture of cement, fertilizer, rayon, paper pulp, chemicals and pharmaceuticals.

(x) Sulphur

- Sulphur is used in chemical industries to manufacture sulphuric acid, paints, explosive materials, rayon and fertilizers.

(i) Chromite

- Chromite gives hardness and electrical resistance to steel.
- It is used for bridges and railways carriages.
- It is also used in metallurgical furnaces and for making engineering tools and stainless steel etc.

(ii) Iron ore

- Steel making, construction and the transport industry.

(iii) Copper

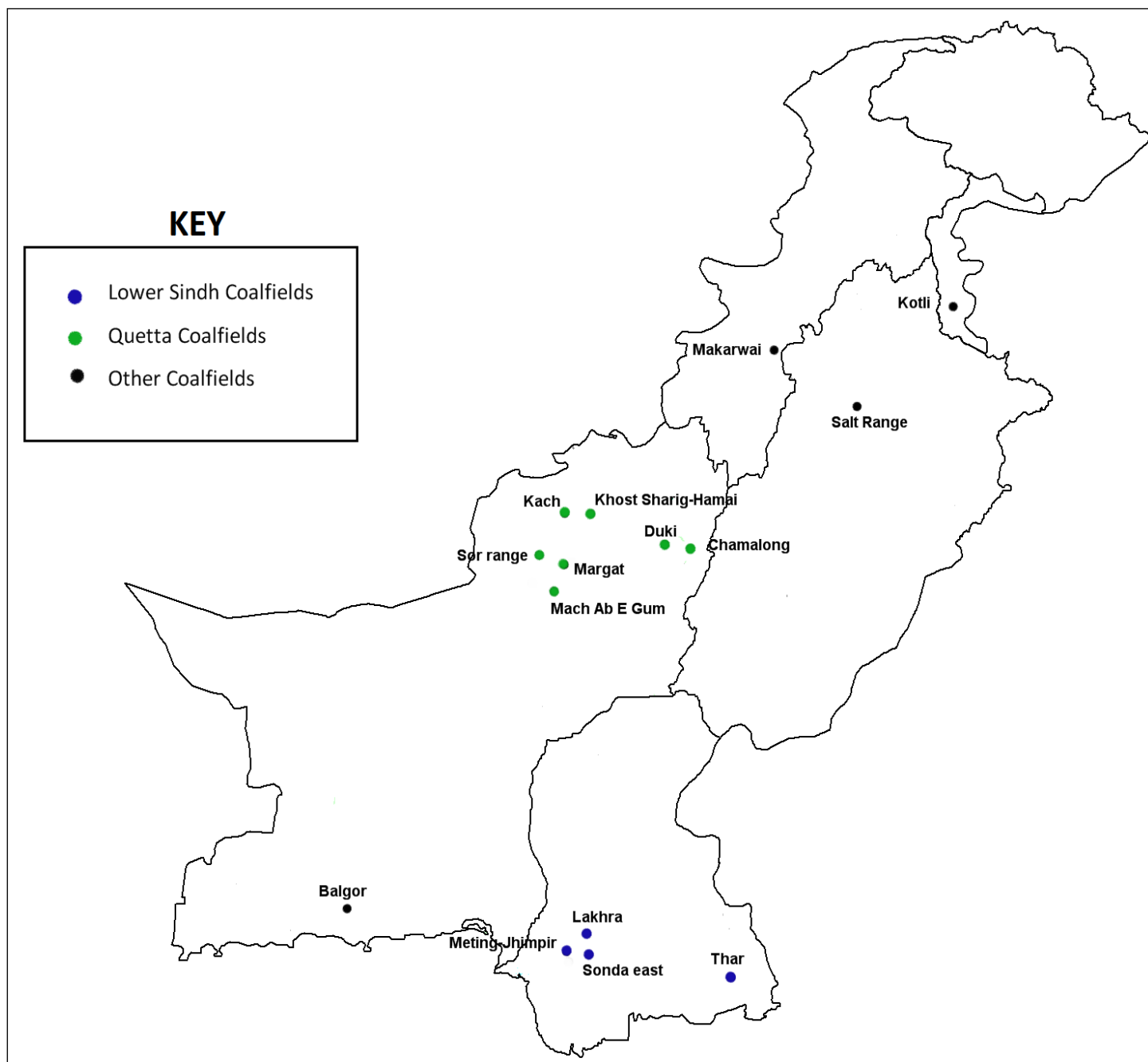
- Making electrical wires and other electrical appliances especially switches that carry current.
- It is also used in making alloys, water pipes and tanks.

(iv) Manganese

- Used in making dry batteries and paints.
- It is vital alloy in steel making.

(v) Bauxite:

- It is used in utensils, tins, cans etc.



KEY



Gas fields



Oil fields



Oil and Gas fields

